

Optimizing Ambulance Locations for Maximized Survival Rates in Baltimore

Ayesha Khan

Department of Machine Learning
University of Maryland
College Park, MD, USA

Yaswanth Kottana

Department of Machine Learning
University of Maryland
College Park, MD, USA

Abstract—The emergency medical response time is a primary determinant of patient survival. We formulate ambulance positioning in Baltimore, Maryland, as a Mixed Integer Linear Program (MILP) that maximizes expected patient survival across the city using the clinically validated Larsen et al. (1993) survival function. A K-means clustering step applied to 80,959 historical 911 calls produces 100 demand zones with call-volume weights, which directly inform the optimization. Candidate sites are the 58 official Baltimore City career fire stations. A demand cap constraint grounded in the EMS Unit Hour Utilization (UHU) standards prevents the overloading of the ambulance. We evaluate performance across fleet sizes $P \in \{23, \dots, 58\}$ and compare against a random naive baseline of 20-trials. Our optimizer achieves a city-wide survival rate of 60.83% with $P=25$ ambulances (+1.59% over naive) and 61.39% at $P=45$, where the system saturates — adding further ambulances yields zero marginal benefit. At $P=35$ all 100 demand zones achieve coverage within 8 minutes. Minimum feasibility analysis shows at least 23 ambulances are required to satisfy the demand cap constraint.

Index Terms—ambulance positioning, facility location, mixed integer programming, emergency medical services, survival optimization, K-means clustering, Baltimore

I. INTRODUCTION

Emergency medical response is among the most time-critical public services a city provides. For out-of-hospital cardiac arrest, survival probability decreases by approximately 2.3% for every minute of delayed response [1]. Despite its importance, the strategic placement of ambulances — determining *where* to station units when not on a call — remains an open operational challenge in many cities.

Traditional approaches rely on uniform geographic coverage or simple heuristics based on historical call density [3]. These methods fail to encode the explicit relationship between response time and patient outcomes. More principled work frames the problem as a facility location optimization [2], [4], but typically uses binary coverage or distance as proxies rather than clinical survival probability.

This paper makes the following contributions:

- We formulate ambulance positioning as a survival-maximizing MILP, embedding the Larsen et al. survival function directly in the objective.
- We integrate K-means demand estimation on real Baltimore 911 call data, connecting the ML preprocessing step to the optimization objective through call-volume weights.

- We introduce a time-weighted demand cap derived from EMS UHU standards, preventing ambulance overload.
- We identify a saturation threshold of $P=45$ — the minimum fleet size beyond which no additional survival benefit is achievable given Baltimore’s 58-station network.
- We benchmark against a 20-trial averaged random baseline across all fleet sizes, demonstrating consistent improvement from optimization.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the mathematical formulation. Section IV describes data and methodology. Section V presents numerical results. Section VI discusses challenges and lessons learned. Section VII concludes.

II. RELATED WORK

The ambulance location problem has a rich history in operations research. The p-median model [3] minimizes weighted average distance from demand points to facility locations. While tractable and widely deployed, it uses distance as a proxy for outcome, ignoring the nonlinear relationship between response time and survival.

Church and ReVelle [2] introduced the Maximum Covering Location Problem (MCLP), which maximizes demand covered within a threshold distance. Our formulation generalizes MCLP by replacing the binary coverage indicator with a continuous survival probability function, eliminating the need for an arbitrary coverage radius. Daskin [4] extended coverage models to account for probabilistic ambulance availability; our demand cap constraint addresses this through UHU-based capacity limits.

The Larsen et al. [1] linear survival approximation, calibrated on cardiac arrest outcomes, has been operationalized in planning tools but rarely embedded directly as an optimization objective. Brotcorne et al. [5] provide a comprehensive survey of ambulance location and relocation models.

III. PROBLEM FORMULATION

A. Decision Variables

Let J denote the set of 58 official Baltimore career fire stations and I the set of 100 demand zones from K-means. We define:

- $x_j \in \{0, 1\}$: 1 if an ambulance is stationed at site j .

- $y_{ij} \in [0, 1]$: fraction of zone i 's demand served by station j . Continuous relaxation enables demand splitting (MILP structure).

B. Parameters

- w_i : aggregate call volume at zone i .
- t_{ij} : mean travel time (min) from station j to zone i , averaged across all individual call locations within the zone:

$$t_{ij} = \frac{d(i, j) \times 1.33}{0.5} + 1.0 \quad (1)$$

where $d(i, j)$ is haversine distance (miles), 1.33 is the Baltimore circuitry factor [6], $v=0.5$ miles/min, $T_{\text{dispatch}}=1$ min.

- p : fleet size (decision parameter swept from 23 to 58).
- $P(S | t_{ij}) = 0.67 - 0.023 \cdot t_{ij}$: Larsen survival function [1]. Pairs with $t_{ij} > 29$ min excluded.
- $W_{\max} = 210,240$ call-min/year: capacity per station ($\text{UHU} = 0.40$, 45 min task time).

C. Objective Function

$$Z = \sum_{i \in I} \sum_{j \in J} w_i \cdot (0.67 - 0.023 \cdot t_{ij}) \cdot y_{ij} \quad (2)$$

Each term encodes call frequency (w_i), clinical survival probability, and assignment fraction simultaneously. This generalizes MCLP [2] by replacing binary coverage with a continuous survival function.

D. Constraints

C1 — Resource Limit:

$$\sum_{j \in J} x_j = p \quad (3)$$

C2 — Full Coverage:

$$\sum_{j \in J} y_{ij} = 1 \quad \forall i \in I \quad (4)$$

C3 — Existence Link:

$$y_{ij} \leq x_j \quad \forall i \in I, j \in J \quad (5)$$

C4 — Integrality:

$$x_j \in \{0, 1\}, \quad y_{ij} \in [0, 1] \quad \forall i, j \quad (6)$$

C5 — Demand Cap:

$$\sum_{i \in I} w_i \cdot y_{ij} \cdot (t_{ij} + 45) \leq W_{\max} \cdot x_j \quad \forall j \in J \quad (7)$$

Adding 45 minutes on-scene time captures full task duration. Feasibility requires $p \cdot W_{\max} \geq \sum_i w_i$, satisfied when $p \geq 23$ (empirically: $p_{\min} = 22.7$).

TABLE I
BALTIMORE 911 DATASET SUMMARY

Attribute	Value
Total raw records	432,766
After non-emergency filter	96,799
After geocoding (83.6%)	80,959
Date range	Jan 1 – Mar 16, 2026
Demand zones ($ I $)	100
Candidate stations ($ J $)	58
$W_{\max}$	210,240 call-min/year
Min. feasible fleet	$22.7 \approx 23$

IV. DATA AND METHODOLOGY

A. Dataset

We use Baltimore City's 911 call-for-service dataset (data.baltimorecity.gov) covering January 1 through March 16, 2026. After filtering out non-emergency calls and geocoding, 80,959 emergency calls remain (83.6% geocoding success from 96,799 priority calls; 432,766 total raw records). Table I summarizes the dataset.

B. Demand Estimation

K-means ($k=100$, `random_state=42`) clusters geocoded calls. Each cluster i forms a demand zone with weight w_i equal to its call count. t_{ij} is the mean travel time from all individual call locations within zone i to station j , rather than the centroid-to-centroid distance. The resulting (100×58) matrices yield a mean survival probability of 0.3383 before optimization. Fig. 1 shows demand zones sized by w_i .

C. Solver and Baseline

The MILP is solved with PuLP/CBC (no commercial license). Solve times range from 0.23 to 2.14 seconds across all fleet sizes.

The **naive baseline** randomly selects p stations (averaged over 20 seeds), assigns each zone to its nearest open station, and reports the resulting survival rate — providing a same-resource comparison for each fleet size.

V. NUMERICAL RESULTS

A. Optimal Placement

Fig. 2 shows the $P=25$ optimized placement. Stations concentrate in central and west Baltimore, co-located with high- w_i zones. Several peripheral stations are bypassed by the optimizer, as they cannot improve the weighted survival objective under C5.

B. Sensitivity Analysis

Table II and Fig. 3 present full results. Four key findings emerge.

Saturation at $P = 45$. Both survival rate and response time plateau completely at $P=45$. Ambulances 46–58 yield exactly zero marginal survivors. This follows directly from C5: once 45 stations collectively satisfy all demand within the

Baltimore 911 Demand Zones: Sized by Call Volume (w_i)

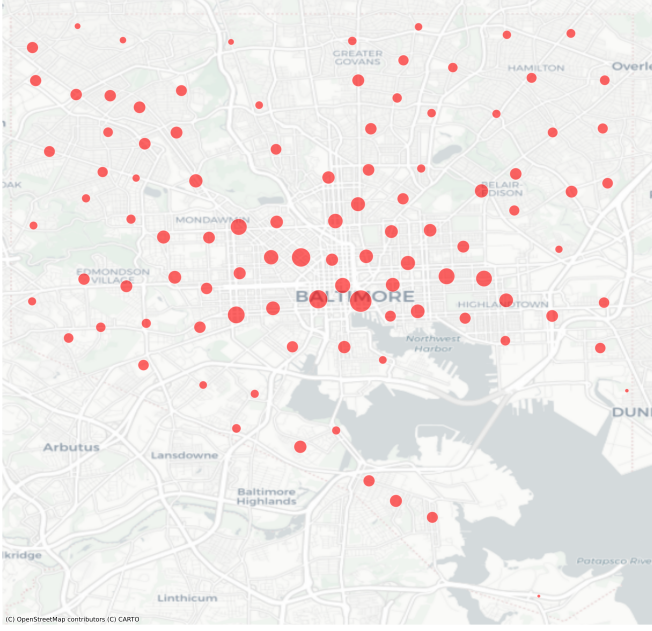


Fig. 1. Baltimore 911 demand zones from K-means ($k=100$). Circle size proportional to w_i . High-demand zones concentrate in central and west Baltimore.

Baltimore 911 Demand Zones, All Fire Stations, and Optimal 25 Ambulance Stations

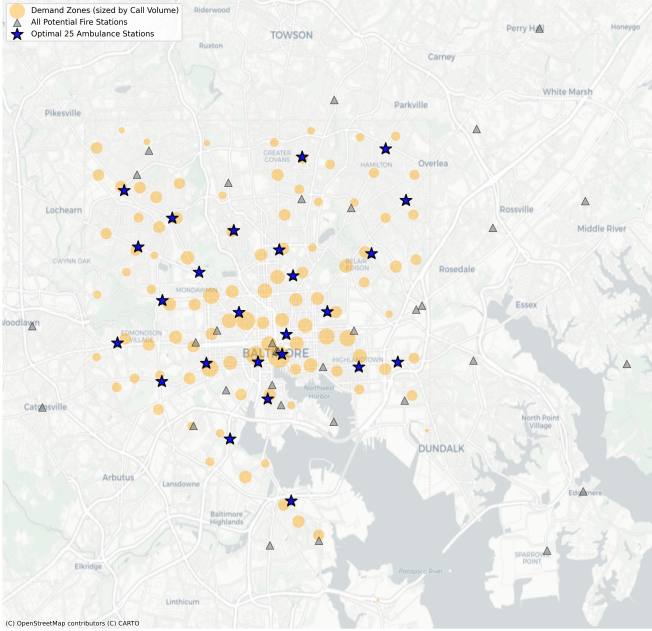


Fig. 2. Optimal 25-ambulance placement (blue stars) over demand zones (orange circles, sized by w_i) and all 58 candidate stations (grey triangles). Stations cluster near high-demand areas.

UHU capacity limit, additional stations have no assignable demand. Practically, Baltimore’s 58-station network achieves its theoretical maximum with 45 ambulances — a 22.4% reduction from full deployment with no survival penalty.

TABLE II
OPTIMIZATION RESULTS VS. NAIVE BASELINE

P	Survival Rate	Survivors /period	Avg Time (min)	Zones ≤ 8 min	vs Naive
23	60.67%	49,116	2.75	99/100	+2.52%
24	60.76%	49,192	2.71	99/100	+1.64%
25	60.83%	49,251	2.68	99/100	+1.59%
35	61.30%	49,630	2.48	100/100	+1.14%
36	61.32%	49,647	2.47	100/100	+1.16%
45	61.39%	49,703	2.44	100/100	+0.89%
47	61.39%	49,703	2.44	100/100	+0.70%
50	61.39%	49,703	2.44	100/100	+0.38%
58	61.39%	49,703	2.44	100/100	−0.00%

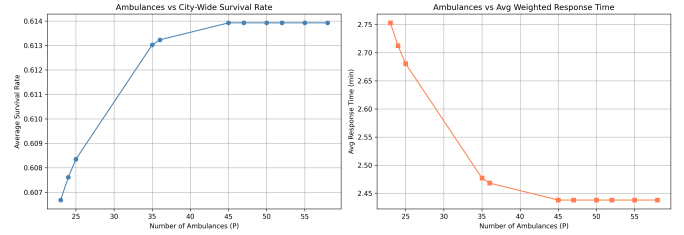


Fig. 3. Sensitivity across $P \in \{23, \dots, 58\}$. *Left*: survival rate rises steeply from $P=23$ to $P=35$, plateaus at $P=45$. *Right*: weighted response time falls from 2.75 to 2.44 min, also saturating at $P=45$.

TABLE III
MARGINAL EXPECTED SURVIVORS PER ADDITIONAL AMBULANCE

Transition	Units Added	Marginal Survivors/Unit
23 $\rightarrow$ 24	1	75.29
24 $\rightarrow$ 25	1	59.69
25 $\rightarrow$ 35	10	37.85
35 $\rightarrow$ 36	1	16.61
36 $\rightarrow$ 45	9	6.24
45 $\rightarrow$ 58	13	0.00

Minimum feasibility at $P = 23$. Feasibility analysis yields $p_{\min} = 22.7$; below 23 ambulances no assignment satisfying C5 exists.

Steep marginal returns at low fleet sizes. Table III shows the marginal survivor analysis. The 24th ambulance saves 75.29 additional expected lives; the 25th saves 59.69. Each of the 10 ambulances added from 25 to 35 saves 37.85 on average. Beyond 36 the rate falls sharply to 6.24 (ambulances 36–45) and then zero.

Consistent improvement over naive baseline. The optimizer outperforms random placement at every fleet size from $P=23$ through $P=55$. The largest gain is at $P=23$ (+2.52%), where intelligent placement under tight capacity matters most. The naive baseline with $P=23$ achieves 58.54% survival at 3.68 min average response time, versus 60.67% and 2.75 min for the optimized solution — a difference of approximately 1,300 additional expected survivors over the data period with

the same ambulance budget.

VI. DISCUSSION AND CHALLENGES

A. Challenges Encountered

Formulation evolution. The initial formulation assigned exactly one ambulance per zone with binary y_{ij} . The professor identified that a single ambulance cannot realistically serve an entire high-demand neighborhood. Two changes followed: relaxing y_{ij} to continuous $[0, 1]$ (enabling demand splitting), and introducing C5.

Objective function double-counting. An early version used x_j rather than y_{ij} in the objective, causing every placed ambulance to receive survival credit for every zone simultaneously. Correcting this required cleanly separating placement (x_j) from assignment (y_{ij}).

Mean travel time approximation. The professor raised the concern that centroid-to-centroid haversine distance collapses spatially distributed calls to a single point. We addressed this by computing the mean t_{ij} across all individual call locations within each zone — a direct implementation of the feedback.

Unit consistency in C5. Adding t_{ij} to C5 changed units to call-minutes, requiring $W_{\max}$ re-derivation: $8,760 \times 0.40 \times 60 = 210,240$ call-min/year.

Integer programming complexity. The MILP requires branch-and-bound solvers rather than the gradient-based methods covered in class. Despite theoretical NP-hardness, CBC solves all instances in under 2.2 seconds at our scale (100 zones, 58 stations), confirming that NP-hardness is a worst-case theoretical concern, not a practical one at this problem size.

B. Limitations

- **Partial-year data.** The dataset covers only January–March 2026 (~75 days). Demand weights w_i reflect winter patterns. A full-year dataset would improve generalizability.
- **Static model.** Ambulances are assumed always available. Stochastic availability (busy probability) would require dynamic simulation.
- **Larsen generalization.** The survival function was calibrated on cardiac arrest data. Different emergency types have different time-outcome relationships.
- **Road network distance.** Despite the 1.33 circuitry factor, haversine remains an approximation; OpenStreetMap routing would improve accuracy.

C. Key Observation: Policy Implication of Saturation

The saturation at $P=45$ is the most practically significant finding. It reveals that Baltimore’s station network, used optimally, achieves maximum theoretical survival with 13 fewer ambulances than full deployment. Combined with the steep marginal return curve, the data suggest that the most cost-effective EMS investment is in the 23–36 ambulance range, where each additional unit saves 16–75 expected lives per data period.

VII. CONCLUSION

We presented a survival-maximizing MILP for ambulance positioning in Baltimore, Maryland. By directly embedding the Larsen et al. clinical survival function in the objective and grounding the demand cap in EMS operational standards, the formulation optimizes for patient outcomes rather than proxy metrics. Applied to 80,959 real 911 calls, the optimizer achieves 60.83% city-wide survival at $P=25$ (+1.59% over naive, $\approx 1,300$ additional expected survivors) and reaches its theoretical maximum of 61.39% at $P=45$, beyond which no marginal improvement is achievable given the station network. Full 8-minute zone coverage is obtained at $P=35$.

The primary lesson from this project is that real optimization problems require iterative formulation. Our constraints and objective evolved substantially through advisor feedback, and several structural errors were corrected before implementation. Every simplification must be explicitly justified and acknowledged.

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